# JEE Mathematics: Quick Review with Grading and Practice Map - Continuity & Differentiability (Batch XII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (64 questions; a question that appears in two of these is counted once): High = 6 or more questions, Medium = 1 to 5 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

## Definition of Continuity

### Question 01

What is the graphical definition of continuity?

**Model Answer**

$f$ is continuous at $x = a$ if its graph can be drawn through $x = a$ without lifting the pen (no break, no hole, no jump).

Ex: $y = \sin x,\ \cos x,\ |x|$ are continuous everywhere; $y = [x]$ and $y = \{x\}$ break at every $x \in I$.

**Grade - Low**

Practice map, main concept: none

Also used in: none

### Question 02

What is the mathematical definition of continuity?

**Model Answer**

$$\lim_{h\to 0} f(a-h) = \lim_{h\to 0} f(a+h) = f(a), \quad h > 0$$

i.e. LHL = RHL = $f(a)$ (all three finite). To find a constant, equate the one-sided limits with $f(a)$.

Ex: $f = x + 1\ (x \le 1),\ 6 - kx^2\ (x > 1)$ continuous at 1: $2 = 6 - k \Rightarrow k = 4$.

**Grade - High**

Practice map, main concept: Sheet: J-I 1; J-II 1-4, 6-9, 12, 38; J-M 1-4, 8, 12-13, 22-24, 27; J-A 1; C-I 2, 4, 8; C-II 2, 5-6, 8, 15, 22 | DPP: 1(1-3, 5-10, 12-15, 17-19); 2(1-2, 4, 10, 12-17) | Question Bank: 118-119, 126, 129

Also used in: Sheet: J-I 2-6, 10, 12; J-II 5, 13, 16, 22, 24, 29-30, 32-33, 42, 44, 50, 59; J-M 16, 18; J-A 8; C-I 1, 3, 5-6, 17-18; C-II 3-4, 7 | DPP: 1(4); 2(5-7, 11); 3(1, 3, 6, 8, 10-12); 4(8, 11) | Question Bank: 113

### Question 03

What to keep in mind while checking continuity if one side of the point is not in the domain?

**Model Answer**

Only the side inside the domain is checked: at the end point $a$ (domain on the right) $f$ is continuous if RHL = $f(a)$.

Ex: $\sqrt{x}$ is continuous at 0 (RHL = $0 = f(0)$); $\sin^{-1} x$ is continuous at 1 (LHL = $\frac{\pi}{2} = f(1)$).

**Grade - Medium**

Practice map, main concept: none

Also used in: Sheet: J-I 3 | Question Bank: 117

## Important Points related to Continuity

**Question 04**

What is the relation between existence of limit and continuity at a point?

**Model Answer**

Continuous $\Rightarrow$ limit exists; limit exists $\not\Rightarrow$ continuous. Continuity is discussed only at points of the domain ($a$ must be in the domain).

Ex: $[x] + [-x]$ at 0: limit $= -1$ but $f(0) = 0$, so not continuous. $\frac{1}{x}$ is continuous (in its domain).

**Grade** - **Low**

Practice map, main concept: none

Also used in: DPP: 2(8)

**Question 05**

What is the condition for a function to be left continuous at $x = a$?

**Model Answer**

$$\lim_{x \to a^-} f(x) = f(a)$$

Ex: $[x]$ at $x = 1$: LHL $= 0 \neq f(1) = 1$, so $[x]$ is not left continuous at 1.

**Grade** - **Low**

Practice map, main concept: none

Also used in: none

**Question 06**

What is the condition for a function to be right continuous at $x = a$?

**Model Answer**

$$\lim_{x \to a^+} f(x) = f(a)$$

Ex: $[x]$ at $x = 1$: RHL $= 1 = f(1)$, so $[x]$ is right continuous at every integer.

**Grade** - **Low**

Practice map, main concept: none

Also used in: none

**Question 07**

Comment on continuity of $f + g$, $f - g$, $f \cdot g$, $f/g$ if both $f$ and $g$ are continuous at a point.

**Model Answer**

All four are continuous ($f/g$ needs $g(a) \neq 0$). Polynomial, trigonometric, exponential and logarithmic functions are continuous in their domains.

Ex: $\dfrac{x^2 + 1}{x - 2}$ is continuous for all $x \neq 2$.

**Grade** - **Low**

Practice map, main concept: none

Also used in: Sheet: J-II 48; C-II 1

**Question 08**

Comment on continuity of $f+g$, $f-g$, $f\cdot g$, $f/g$ if both $f$ and $g$ are discontinuous at a point.

**Model Answer**

Nothing definite can be said about any of them; check directly.

Ex: $[x]$, $\{x\}$ are discontinuous at 1, but $[x]+\{x\}=x$ and $[x]\{x\}$ are continuous at 1; $[x]\{x\}$ is discontinuous at 0 (LHL $=-1$, RHL $=0$).

**Grade** - Low

Practice map, main concept: Sheet: J-I 5

Also used in: none

**Question 09**

Comment on continuity of $f+g$, $f-g$, $f\cdot g$, $f/g$ if $f$ is continuous and $g$ is discontinuous at a point.

**Model Answer**

$f\pm g$ is surely discontinuous; nothing definite about $f\cdot g$ and $f/g$.

Ex: $x[x]$ is continuous at $x=0$ (both limits 0) but discontinuous at $x=1$ (LHL $=0$, RHL $=1$).

**Grade** - Low

Practice map, main concept: Sheet: J-I 4

Also used in: DPP: 2(7)

## Continuity in an Interval

**Question 10**

What is the condition for a function to be continuous in $(a,b)$?

**Model Answer**

$f$ is continuous at each and every point of $(a,b)$.

Ex: $\frac{1}{x}$ is continuous in $(0,1)$ although it is unbounded there.

**Grade** - Medium

Practice map, main concept: Sheet: J-A 14

Also used in: none

**Question 11**

What is the condition for a function to be continuous in $[a,b]$?

**Model Answer**

(i) continuous in $(a,b)$, (ii) right continuous at $a$, (iii) left continuous at $b$.

Ex: $[x]$ is not continuous in $[0,1]$ (not left continuous at 1); $\sqrt{1-x^2}$ is continuous in $[-1,1]$.

**Grade** - Low

Practice map, main concept: Sheet: J-I 14

Also used in: Sheet: J-II 10

**Question 12**

What are the checkpoints for continuity in an interval?

**Model Answer**

(i) points where the definition changes, (ii) for $[f(x)]$, $\{f(x)\}$: points where $f(x)$ is an integer, (iii) for $\operatorname{sgn} f(x)$: points where $f(x) = 0$, (iv) points where a denominator is 0 or the domain breaks.

Ex: $\operatorname{sgn}(x-2) + [x - \frac{1}{2}] + [\ln x]$ on $[1, 3]$ is discontinuous at $x = \frac{3}{2},\ 2,\ \frac{5}{2},\ e$.

**Grade - High**

Practice map, main concept: Sheet: J-I 2, 12; J-II 10-11, 14-15, 39, 48, 50, 54-55; J-M 14, 18, 20; J-A 8, 15; C-II 21 | DPP: 1(4); 2(3, 6-8) | Question Bank: 117, 125, 128

Also used in: Sheet: J-I 14; J-II 2, 27, 31, 37, 45, 59-61; J-M 7, 9-10, 15, 19, 21, 25, 29; J-A 7; C-I 15-16, 22; C-II 10, 14 | DPP: 2(1); 3(2, 7); 4(1, 9-10, 13) | Question Bank: 127

**Question 13**

How to check continuity of a function defined separately for rational and irrational numbers?

**Model Answer**

$f = g(x)$ for $x \in Q$, $h(x)$ for $x \notin Q$ ($g, h$ continuous): $f$ is continuous only at the points where $g(x) = h(x)$; discontinuous at all other points.

Ex: $f = x^2 + 6$ $(x \in Q)$, $5x$ $(x \notin Q)$: $x^2 - 5x + 6 = 0$, continuous only at $x = 2, 3$.

**Grade - Low**

Practice map, main concept: Sheet: C-I 9, 12 | DPP: 1(11)

Also used in: Sheet: C-I 1

**Question 14**

How to check continuity of $[f(x)]$ if the graph of $f$ can be drawn?

**Model Answer**

Draw $y = f(x)$ and the lines $y = n$ $(n \in I)$. $[f]$ is discontinuous where the graph crosses an integer line, or touches it from below (local max at an integer value); touching from above (local min) keeps $[f]$ continuous.

Ex: $[\sin x]$ in $[0, \pi]$ is discontinuous only at $x = \frac{\pi}{2}$.

**Grade - Medium**

Practice map, main concept: Sheet: C-I 7

Also used in: Sheet: J-II 10, 28; J-M 20, 28; J-A 7, 9; C-II 7, 10, 21, 23 | DPP: 2(3); 4(9)

**Question 15**

How to find continuity of functions with built-in limit?

**Model Answer**

First evaluate the limit separately for $|x| < 1$, $|x| = 1$, $|x| > 1$ ($x^{2n} \to 0,\ 1,\ \infty$), write $f$ piecewise, then check continuity at $x = \pm 1$.

Ex: $f = \lim_{n\to\infty} \dfrac{x^{2n-1} + ax^2 + bx}{x^{2n} + 1}$: $ax^2 + bx$ $(|x| < 1)$, $\frac{1}{x}$ $(|x| > 1)$; continuous on $R$ iff $a = 0,\ b = 1$.

**Grade - Medium**

Practice map, main concept: Sheet: J-II 13; J-M 25; C-I 1, 5, 17; C-II 1, 3-4, 7

Also used in: none

## Types of Discontinuity

**Question 16**

What is the condition for removable type of discontinuity?

**Model Answer**

$\lim_{x\to a} f(x)$ exists (finite) but $\neq f(a)$ or $f(a)$ is not defined. Redefine $f(a) = \lim_{x\to a} f(x)$ to make $f$ continuous.

Ex: $\dfrac{x^2-4}{x-2}$ at $x = 2$: limit 4; define $f(2) = 4$.

**Grade** - **Low**

Practice map, main concept: Sheet: J-I 6; J-II 46; C-I 18

Also used in: Sheet: J-II 4, 49, 60 | DPP: 1(9); 2(2, 10, 17)

**Question 17**

What is the condition for non-removable type of discontinuity?

**Model Answer**

$\lim_{x\to a} f(x)$ does not exist (LHL $\neq$ RHL, or a one-sided limit is infinite or oscillates); no redefinition of $f(a)$ can help.

Ex: $[x]$ at $x = 1$ (LHL $= 0$, RHL $= 1$).

**Grade** - **Low**

Practice map, main concept: Sheet: J-II 5, 17, 49; C-I 3

Also used in: Sheet: J-II 46; C-II 19

**Question 18**

What are the different types of removable discontinuity and the condition for each?

**Model Answer**

Missing point: limit exists, $f(a)$ not defined. Isolated point: limit exists, $f(a)$ defined but $\neq$ limit.

Ex: $\frac{\sin x}{x}$ at 0: missing point. $[x] + [-x]$ at 0: isolated point (limit $-1$, $f(0) = 0$).

**Grade** - **Low**

Practice map, main concept: none

Also used in: Sheet: J-II 46; C-I 18

**Question 19**

What are the different types of non-removable discontinuity and the condition for each?

**Model Answer**

Finite (jump): both one-sided limits finite and unequal. Infinite: at least one one-sided limit is $\pm\infty$. Oscillatory: the limit oscillates between two finite values.

Ex: $\tan^{-1}\frac{1}{x}$ at 0: $\pm\frac{\pi}{2}$, finite; $\frac{1}{x^2}$ at 0: infinite; $\sin\frac{1}{x}$ at 0: oscillatory.

**Grade** - **Low**

Practice map, main concept: Sheet: C-II 19 | DPP: 2(11)

Also used in: Sheet: J-II 17, 49, 53 | DPP: 2(9, 13-15)

**Question 20**
What is the jump of discontinuity in finite type discontinuity?

**Model Answer**

$$\text{Jump} = |\text{RHL} - \text{LHL}|$$

Ex: $\frac{|\sin x|}{x}$ at 0: $\pm 1$, jump 2; $\frac{[x]}{x}$ at 2: 1 and $\frac{1}{2}$, jump $\frac{1}{2}$.

**Grade** - Low
Practice map, main concept: DPP: 2(9)
Also used in: Sheet: J-II 53; C-II 19 | DPP: 2(11)

**Question 21**
What are the checkpoints for continuity of $f(g(x))$ if the checkpoints for $f$ and $g$ are known?

**Model Answer**
If $f$ is discontinuous at $x = a, b$ and $g$ at $x = c, d$: check $f(g(x))$ at $x = c, d$ and at all $x$ with $g(x) = a$ or $g(x) = b$. (For $f \pm g$: $a, b, c, d$ are all discontinuities; for $fg$, $f/g$ they are checkpoints.)
Ex: $f(x) = \frac{1}{1-x}$: $f(f(x)) = \frac{x-1}{x}$ is discontinuous at $x = 1$ and at $x = 0$ (where $f(x) = 1$).

**Grade** - Medium
Practice map, main concept: Sheet: J-I 3; J-II 61; J-M 19, 21; C-II 11, 14 | DPP: 2(5, 18)
Also used in: Sheet: J-II 48, 56; J-M 23; J-A 13; C-I 4 | DPP: 1(11)

**Question 22**
State the extreme value theorem.

**Model Answer**
If $f$ is continuous on $[a, b]$, then $f$ attains a least value $m$ and a greatest value $M$ on $[a, b]$ (so $f$ is bounded and its range is $[m, M]$).
Ex: $x^2$ on $[-1, 2]$: $m = 0$, $M = 4$. On $(0, 1]$, $\frac{1}{x}$ has no greatest value (interval not closed).

**Grade** - Low
Practice map, main concept: Sheet: C-I 19
Also used in: Sheet: C-I 1

## Intermediate Value Theorem (IMVT)

**Question 23**
Write the statement of IMVT.

**Model Answer**
If $f$ is continuous on $[a, b]$ and $f(a) \neq f(b)$, then for every $c$ between $f(a)$ and $f(b)$ there is at least one $x_0 \in (a, b)$ with $f(x_0) = c$.

**Grade** - Low
Practice map, main concept: none
Also used in: Sheet: C-I 14

**Question 24**
Write the alternate statement of IMVT.

**Model Answer**
If $f$ is continuous on $[a,b]$ and $f(a)f(b)<0$, then $f(x)=0$ has at least one root in $(a,b)$ (there may be more). The condition is sufficient, not necessary.
Ex: $x^3+x-1$: $f(0)=-1$, $f(1)=1$, so a root lies in $(0,1)$.

**Grade** - **Low**
Practice map, main concept: Sheet: J-II 18, 47; C-I 14
Also used in: Sheet: C-I 19 | DPP: 4(5)

**Question 25**
How to determine whether a question is of IMVT?

**Model Answer**
Only continuity and end values are given and the question asks to show that an equation has at least one solution in an interval, or that some $c$ exists with $f(c)=\ldots$.

**Grade** - **Low**
Practice map, main concept: none
Also used in: Sheet: C-I 14

**Question 26**
How to apply IMVT in questions?

**Model Answer**
Bring all terms to one side: $g(x)=\text{LHS}-\text{RHS}$; check $g$ is continuous on $[a,b]$; show $g(a)g(b)\le 0$ (for $f(c)=c$ take $g=f(x)-x$; for $f(c)=f(c+k)$ take $g=f(x+k)-f(x)$).
Ex: $(x-1)(x-3)=(x-2)(x-4)$: $g(2)=-1$, $g(4)=3$, so a root lies in $(2,4)$.

**Grade** - **Low**
Practice map, main concept: Sheet: J-II 19-20 | DPP: 1(16); 4(5)
Also used in: Sheet: J-II 18, 47

## Definition of Differentiability

**Question 27**
What is the relation between the tangent at a point and derivability of the function at that point?

**Model Answer**
$f$ is derivable at $x=a$ iff a unique tangent of finite slope exists there. A vertical tangent ($x^{1/3}$ at 0) or no unique tangent ($|x|$ at 0) means not derivable.

**Grade** - **Low**
Practice map, main concept: none
Also used in: Sheet: C-I 10

**Question 28**
Write the expressions of LHD and RHD.

**Model Answer**

$$\text{RHD}=f'(a^+)=\lim_{h\to 0}\frac{f(a+h)-f(a)}{h},\qquad \text{LHD}=f'(a^-)=\lim_{h\to 0}\frac{f(a-h)-f(a)}{-h},\quad h>0$$

Each must exist and be finite.
Ex: $|x|$ at 0: LHD $=-1$, RHD $=1$.

**Grade - Medium**

Practice map, main concept: Sheet: J-I 9; J-II 23, 26, 30; C-I 6

Also used in: Sheet: J-I 8; J-II 22, 58; J-A 2, 9, 15, 17; C-II 12 | DPP: 3(10); 4(7)

### Question 29

Write all the different ways in which LHD and RHD can be written.

**Model Answer**

$$\text{RHD} = \lim_{x\to a^+} \frac{f(x)-f(a)}{x-a} = \lim_{h\to 0^+} \frac{f(a+h)-f(a)}{h}, \qquad \text{LHD} = \lim_{x\to a^-} \frac{f(x)-f(a)}{x-a} = \lim_{h\to 0^+} \frac{f(a)-f(a-h)}{h}$$

If $f$ is derivable at $a$: $\lim_{h\to 0} \dfrac{f(a+ph)-f(a+qh)}{h} = (p-q)f'(a)$.

Ex: $f'(2) = 3 \Rightarrow \lim_{h\to 0} \dfrac{f(2+h)-f(2-h)}{h} = 6$.

**Grade - Medium**

Practice map, main concept: Sheet: J-I 15, 22-23; J-II 43, 51, 58; J-A 10; C-I 10; C-II 9, 13 | DPP: 3(4); 4(6, 8)

Also used in: Sheet: J-II 32-33, 35-36, 44; J-A 5, 11-12; C-I 11 | Question Bank: 130

### Question 30

What is the condition for a function to be derivable at $x = a$?

**Model Answer**

$$\text{LHD} = \text{RHD} = \text{finite} = f'(a)$$

For unknown constants: first make $f$ continuous at $a$, then equate LHD and RHD.

Ex: $f = ax + b$ $(x \le -1)$, $ax^5 + x + 2b$ $(x > -1)$ derivable: $b = 1$, $a = 5a + 1 \Rightarrow a = -\frac{1}{4}$.

**Grade - Medium**

Practice map, main concept: Sheet: J-I 10; J-II 24, 29, 32-33, 44, 59 | DPP: 3(3, 6, 8, 10, 12); 4(7, 11)

Also used in: Sheet: J-II 21, 52, 56, 60; J-M 16; C-I 8, 22

### Question 31

What to keep in mind while checking derivability if one side of the point is not in the domain?

**Model Answer**

Only the one-sided derivative from the domain side is needed: at a left end point check RHD only (finite $\Rightarrow$ derivable there).

Ex: $\sqrt{x}$ at 0: RHD $= \lim \frac{1}{\sqrt{h}} = \infty$, not derivable; $x^{3/2}$ at 0: RHD $= 0$, derivable.

**Grade - Low**

Practice map, main concept: none

Also used in: none

### Question 32

What is the relation between continuity and derivability at a point?

**Model Answer**

Derivable $\Rightarrow$ continuous; continuous $\not\Rightarrow$ derivable; discontinuous $\Rightarrow$ not derivable; not derivable $\not\Rightarrow$ discontinuous.

Ex: $|x|$ is continuous but not derivable at 0.

**Grade - Medium**

Practice map, main concept: Sheet: J-II 31, 60

Also used in: Sheet: J-I 23; J-II 26, 37, 40, 45, 54; J-M 28-29; J-A 3 | DPP: 3(3)

**Question 33**
What is the condition for a function to be left derivable at $x = a$?

**Model Answer**
LHD at $x = a$ exists and is a finite number.
Ex: $[x]$ at 1: LHD $= \lim \frac{0-1}{-h} = \infty$, not left derivable.

**Grade** - Low
Practice map, main concept: none
Also used in: none

**Question 34**
What is the condition for a function to be right derivable at $x = a$?

**Model Answer**
RHD at $x = a$ exists and is a finite number.
Ex: $[x]$ at 1: RHD $= \lim \frac{1-1}{h} = 0$, right derivable.

**Grade** - Low
Practice map, main concept: none
Also used in: none

**Question 35**
Comment on derivability of $f + g$, $f - g$, $f \cdot g$, $f/g$ if both $f$ and $g$ are derivable at a point.

**Model Answer**
All four are derivable ($f/g$ needs $g(a) \neq 0$). Polynomial, trigonometric, exponential and logarithmic functions are derivable in their domains.

**Grade** - Low
Practice map, main concept: none
Also used in: DPP: 4(3)

**Question 36**
Comment on derivability of $f + g$, $f - g$, $f \cdot g$, $f/g$ if both $f$ and $g$ are non-derivable at a point.

**Model Answer**
Nothing definite; check directly.
Ex: at 0, $|x| + (-|x|) = 0$ is derivable but $|x| + |x|$ is not.

**Grade** - Low
Practice map, main concept: none
Also used in: none

**Question 37**
Comment on derivability of $f + g$, $f - g$, $f \cdot g$, $f/g$ if $f$ is derivable and $g$ is non-derivable at a point.

**Model Answer**
$f \pm g$ is surely non-derivable; nothing definite about $f \cdot g$, $f/g$.
Ex: $x|x|$ is derivable at 0, but $(x + 1)|x|$ is not (LHD $= -1$, RHD $= 1$).

**Grade** - Medium
Practice map, main concept: Sheet: J-A 11
Also used in: none

**Question 38**
What is the condition for a function to be derivable in $(a, b)$?

**Model Answer**
$f$ is derivable at each and every point of $(a, b)$.

**Grade** - Medium
Practice map, main concept: Sheet: C-II 16
Also used in: Sheet: J-II 55; J-A 14

**Question 39**
What is the condition for a function to be derivable in $[a, b]$?

**Model Answer**
(i) derivable in $(a, b)$, (ii) right derivable at $a$, (iii) left derivable at $b$.

**Grade** - Low
Practice map, main concept: none
Also used in: none

**Question 40**
What are the checkpoints for derivability in an interval?

**Model Answer**
(i) all checkpoints of continuity (discontinuous there $\Rightarrow$ not derivable), (ii) for $|f(x)|$: points where $f(x) = 0$, (iii) points where a piece has a vertical tangent, e.g. $(x - a)^{1/3}$.
Ex: $|x - 1| + |x - 2| + |x - 3|$ is not derivable at $x = 1, 2, 3$.

**Grade** - High
Practice map, main concept: Sheet: J-I 11, 13; J-II 27-28, 37, 40, 57; J-M 15, 28-29; J-A 7; C-I 16, 23 | DPP: 3(2, 7, 14); 4(9) | Question Bank: 127
Also used in: Sheet: J-I 17; J-M 11, 26; J-A 4, 6, 13, 15; C-I 7, 13, 21; C-II 11, 21 | DPP: 4(10) | Question Bank: 123

**Question 41**
How to check derivability of a function defined separately for rational and irrational numbers?

**Model Answer**
$f = g(x)$ $(x \in Q)$, $h(x)$ $(x \notin Q)$ can be derivable only at a point $a$ where $g(a) = h(a)$ and $g'(a) = h'(a)$; check there by the definition.
Ex: $x^2$ $(Q)$, $0$ $(Q')$ is derivable at $0$ $(f'(0) = 0)$; $x$ $(Q)$, $0$ $(Q')$ is continuous but not derivable at $0$.

**Grade** - Low
Practice map, main concept: none
Also used in: Sheet: C-I 12

**Question 42**
Comment on continuity and differentiability of $x^p \sin \frac{1}{x}$, $x^p \cos \frac{1}{x}$ at $x = 0$ (value 0 at $x = 0$).

**Model Answer**
Continuous at 0 iff $p > 0$; derivable at 0 iff $p > 1$ $(f'(0) = 0)$; $f'$ continuous at 0 iff $p > 2$.
Ex: $x^2 \sin \frac{1}{x}$ is derivable at 0 with $f'(0) = 0$, but $f'(x) = 2x \sin \frac{1}{x} - \cos \frac{1}{x}$ is not continuous at 0.

**Grade** - Medium
Practice map, main concept: Sheet: J-I 8; J-II 21; J-A 9, 16-17; C-I 20; C-II 12
Also used in: Sheet: J-A 2

## Concept of LHD and RHD

**Question 43**

What can you say about continuity from the values of LHD and RHD?

**Model Answer**

LHD = RHD = finite: derivable and continuous. Both finite but unequal: continuous, not derivable. At least one infinite: not derivable, continuity cannot be decided.

Ex: $x^{1/3}$ at 0 (both $\infty$) is continuous; $\operatorname{sgn} x$ at 0 (both $\infty$) is discontinuous.

**Grade** - Low

Practice map, main concept: DPP: 3(5)

Also used in: none

**Question 44**

What is the condition for a sharp corner to exist in the graph of a function?

**Model Answer**

$f$ continuous at $a$ and LHD $\neq$ RHD (no unique tangent). Sharp corner $\Rightarrow$ not derivable, but not derivable $\not\Rightarrow$ sharp corner (vertical tangent of $x^{1/3}$).

Ex: $|x|$ and $|x|^{1/3}$ have a sharp corner at 0. For max/min functions, count the corners and breaks on the graph.

**Grade** - High

Practice map, main concept: Sheet: J-I 7; J-II 22, 52; J-M 7, 9-10, 17; J-A 4; C-I 13, 15, 21; C-II 10, 23 | DPP: 4(1-3) | Question Bank: 122-124

Also used in: Sheet: J-II 30, 57-59; C-I 23; C-II 21 | DPP: 3(1, 5, 13-14); 4(12)

## Quick Concept

**Question 45**

How to apply quick concept to check differentiability?

**Model Answer**

Check continuity at $a$ first. Then LHD $= \lim_{x\to a^-} f'(x)$ (derivative of the left piece), RHD $= \lim_{x\to a^+} f'(x)$; compare.

Ex: $e^x - e + 9$ $(x < 1)$, $x^{100} + 7x^{50} + 1$ $(x \geq 1)$: continuous $(9 = 9)$, $f'(1^-) = e$, $f'(1^+) = 450$, not derivable.

**Grade** - High

Practice map, main concept: Sheet: J-II 42, 45, 53 | DPP: 3(1, 11, 13); 4(13)

Also used in: Sheet: J-I 13; J-II 24; J-M 5-6, 15, 17; J-A 3, 16 | DPP: 3(2, 7, 9, 12); 4(11) | Question Bank: 120, 122, 124, 127

**Question 46**

What is the condition to apply quick concept?

**Model Answer**

(i) $f$ continuous at the point, (ii) each piece is easy to differentiate, (iii) $\lim f'(x)$ is not oscillatory.

Ex: not for $x^2 \sin \frac{1}{x}$ at 0: $\lim f'(x)$ oscillates, yet $f'(0) = 0$.

**Grade** - Low

Practice map, main concept: none

Also used in: none

**Question 47**
When will derivability check by quick concept be preferred over the check by basic definition?

**Model Answer**
When $f$ is continuous and the pieces are standard derivable functions (polynomial, exponential, trigonometric). Use the definition for [ ], { }, oscillating terms, functions not given by formulas, or when continuity fails.

**Grade** - Low
Practice map, main concept: none
Also used in: none

**Question 48**
How to find $f'(x)$ when $f(x)$ is a non-uniform function?

**Model Answer**
Differentiate each piece on its open interval; at each junction find LHD and RHD separately and include the point in $f'$ only if they are equal.
Ex: $\frac{x^2}{4} - \frac{3x}{2} + \frac{13}{4}$ $(x < 1)$, $|x-3|$ $(x \geq 1)$: $f' = \frac{x}{2} - \frac{3}{2}$ $(x < 1)$, $-1$ $(1 \leq x < 3)$, $1$ $(x > 3)$; not derivable at 3.

**Grade** - Low
Practice map, main concept: Sheet: J-II 56; C-I 22
Also used in: Sheet: J-II 25, 27, 29, 40, 42, 57; C-I 20

**Question 49**
How to check differentiability of a composite of two functions by defining the composite?

**Model Answer**
Split the domain where the inner function crosses the break points of the outer function; write $f(g(x))$ piecewise on each interval; then check continuity and LHD, RHD at the junctions.
Ex: $f(x) = |x-1|$: $f(f(x)) = ||x-1|-1|$ is not derivable at $x = 0, 1, 2$.

**Grade** - Medium
Practice map, main concept: Sheet: J-II 25; J-M 16; J-A 3, 5, 13 | DPP: 4(10)
Also used in: Sheet: J-II 61; C-II 11, 14

**Question 50**
Comment on differentiability of $g(x)|f(x)|$ at $x = a$ if $f(a) = 0$.

**Model Answer**
($f, g$ derivable, $a$ a simple zero of $f$.) If $g(a) = 0$: derivable at $a$. If $g(a) \neq 0$: not derivable (check). Same for $|f(x)|$ alone ($g = 1$).
Ex: $(x^2-1)|x^2-3x+2|$ is not derivable only at $x = 2$ ($g(1) = 0$, $g(2) = 3$).

**Grade** - High
Practice map, main concept: Sheet: J-I 17; J-M 11, 26; J-A 2, 6 | DPP: 3(9)
Also used in: Sheet: J-A 5, 7, 11; C-II 12

## Defining of Special types of Functions

**Question 51**

How to define $y = \max\{f(t) : a \le t \le x\}$ or $y = \min\{f(t) : a \le t \le x\}$?

**Model Answer**

Sketch $f(t)$ from $t = a$. While $f$ rises (for max) $y = f(x)$; after a peak $y$ stays equal to that peak value until $f$ goes above it again. Write $y$ piecewise, then check.

Ex: $\max\{\sin t : 0 \le t \le x\} = \sin x$ for $0 \le x \le \frac{\pi}{2}$, $= 1$ for $\frac{\pi}{2} \le x \le 2\pi$; continuous at $\frac{\pi}{2}$.

**Grade** - Medium

Practice map, main concept: Sheet: J-II 16; J-M 5-6 | DPP: 4(12) | Question Bank: 120

Also used in: none

## Determination of Function

**Question 52**

What are the steps to determine the function if the functional rule of a differentiable function is given?

**Model Answer**

(1) $f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$; (2) use the rule on $f(x+h) - f(x)$ (find $f(0)$ by $x = y = 0$); (3) get $f'(x)$ in terms of $f'(0)$ and integrate; (4) fix the constant from $f(0)$.

Ex: $f(x+y) = f(x) + f(y) + 2xy - 1$, $f'(0) = 4$: $f(0) = 1$, $f'(x) = 2x + 4$, $f(x) = x^2 + 4x + 1$.

**Grade** - Medium

Practice map, main concept: Sheet: J-I 18-21; J-II 34-36; J-A 12; C-I 11 | DPP: 4(4) | Question Bank: 113, 130

Also used in: none

**Question 53**

How to determine the function by the partial derivative approach if the functional rule of a differentiable function is given?

**Model Answer**

Differentiate the rule with respect to one variable keeping the other constant; then put that other variable $= 0$ (or 1) to get a relation for $f'(x)$; integrate.

Ex: $f(xy) = f(x) + f(y)$, $f'(1) = 3$: w.r.t. $y$: $xf'(xy) = f'(y)$; $y = 1$: $f'(x) = \frac{3}{x}$, $f(x) = 3\ln x$ $(x > 0)$.

**Grade** - Medium

Practice map, main concept: Question Bank: 121

Also used in: none